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Transforming the modelling of nuclear structural materials: A critical review of machine learning approaches

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ABSTRACT

Structural materials in nuclear reactors operate under extreme conditions, including high temperatures, intense irradiation, and corrosive environments. These conditions drive complex microstructural evolution and degradation of mechanical properties. Accurate modelling of these behaviours is essential for predicting component lifetimes and ensuring reactor safety. In recent years, machine learning (ML) has attracted growing interest as a powerful tool for accelerating material modelling and capturing complex environmental effects by exploiting large datasets generated from experiments and simulations. This review critically evaluates the application of ML techniques to the modelling of nuclear structural materials and provides readers with a comprehensive guide to their use in this field. We begin with a conceptual overview of machine learning in material modelling, and highlight key modelling dimensions and representative ML architectures that underpin current applications. The existing literature is then analysed by grouping studies according to material class, while the ML methods employed and their respective application purposes are presented in detail. Finally, ML applications are presented and discussed in relation to the extreme environmental conditions encountered in nuclear systems, followed by a critical evaluation of current challenges and future directions.

1. Introduction

Global energy demand continues to increase rapidly, rising by 2.2% in 2024, a rate that exceeds the average growth of the past decade, driven by increasing electrification, record temperatures, industrial activity, and digitalisation, according to the IEA's Global Energy Review 2025 [1]. Electricity demand alone grew by 4.3%, one of the largest annual increases in recent decades, with emerging economies accounting for the majority of this growth. Meeting this accelerating demand while reducing carbon emissions requires large-scale deployment of reliable, high-output, low-carbon energy sources. Nuclear energy is well positioned to contribute through its reliable baseload operation and absence of operational greenhouse gas emissions [2].

The long-term viability of nuclear energy depends heavily on the performance and reliability of structural materials that must retain their integrity over service life that can extend beyond six decades [3]. These materials form both in-core and out-of-core components, and their performance directly affects reactor safety, efficiency, and overall economic performance. Over their operational lifetime, reactor materials experience extreme combinations of irradiation, temperature, mechanical stress, and corrosive environments, leading to complex degradation mechanisms [4–6].

The selection and qualification of materials for nuclear systems is complicated by the need to balance multiple, often competing considerations, including waste management [7,8] and component performance [9,10]. A wide range of structural and functional materials is used in in-vessel components, each with distinct advantages and limitations [9,11]. For structural materials, the allowable operating temperature window is constrained at the lower end by irradiation-induced embrittlement and at the upper end by thermal creep, both of which can reduce plant thermal efficiency and limit component lifetime [12]. Although the detrimental effects of irradiation are well established, reliable performance data remain sparse and costly to obtain, hindering efforts to establish predictive design margins. Coolant interactions further complicate material behaviour, as corrosion processes can become life-limiting when combined with irradiation and mechanical loading

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List of Machine Learning Abbreviations

ANFIS	Adaptive Neuro-Fuzzy Inference System
ANN	Artificial Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
BI	Bayesian Inference
BO	Bayesian Optimisation
BPNN	Backpropagation Neural Network
CART	Classification and Regression Trees
CNN	Convolutional Neural Network
CondNN	Conditional Neural Network
DT	Decision Tree
DNN	Deep Neural Network
FCNN	Fully Connected Neural Network
GA	Genetic Algorithm
GB	Gradient Boosting
GBR	Gradient Boosting Regression
GKRR	Gaussian Kernel Ridge Regression
GP	Gaussian Process
GPR	Gaussian Process Regression
GPSR	Genetic Programming Symbolic Regression
GRNN	Generalised Regression Neural Network
GRU	Gated Recurrent Unit
KNN	K-Nearest Neighbours
KRR	Kernel Ridge Regression
LLE	Locally Linear Embedding
LLM	Large Language Model
LR	Linear Regression
LSSVM	Least Squares Support Vector Machine
LSTM	Long Short-Term Memory
ML	Machine Learning
MLIP	Machine-Learned Interatomic Potential
NN	Neural Network
PGNN	Physics-Guided Neural Network
PINN	Physics-Informed Neural Network
PSO	Particle Swarm Optimisation
RF	Random Forest
RNN	Recurrent Neural Network
SHAP	SHapley Additive exPlanations
SVR	Support Vector Regression
SVM	Support Vector Machine
TCN	Temporal Convolutional Network
XGBoost	Extreme Gradient Boosting

[13–15]. In liquid-metal systems, where the coolant may also serve as a tritium breeder and neutron multiplier, corrosion can be severe and strongly dependent on temperature and flow rate [16]. These factors illustrate the complexity of predicting nuclear material performance and highlight the need for more advanced modelling approaches.

Predicting the long-term performance of nuclear structural materials has traditionally relied on a combination of atomistic simulations [17,18], mesoscale models [19,20], and continuum-scale constitutive frameworks [21]. While traditional methods have provided essential mechanistic insights, each captures a subset of the relevant physics and often operates on limited length and time scales. Bridging these scales remains challenging, particularly for processes such as irradiation damage accumulation [22,23], corrosion–irradiation synergy [24,25], long-term thermal ageing [26,27], which evolve over decades of operation, and the coupled irradiation and strain-rate-dependent mechanical behaviour [28,29]. Nevertheless, many conventional modelling approaches still depend on empirical parameters derived from sparse and costly experimental data, limiting their ability to generalise beyond the conditions for which they were calibrated. As a result, the predictive capability of conventional modelling approaches remains constrained, especially for next-generation reactor environments that lie outside the historical data envelope.

Machine learning (ML) has emerged as a powerful complementary approach to traditional modelling frameworks, offering the ability to learn complex, nonlinear relationships directly from data generated through experiments and high-fidelity simulations [30,31]. Unlike conventional physics-based models, which require explicit assumptions about governing mechanisms, ML methods can identify patterns across diverse length and time scales and construct predictive models even when mechanistic understanding is incomplete or multiscale coupling is difficult to capture. In materials science and engineering, ML has already accelerated property prediction [32–34], alloy design [35–37], microstructure characterisation [38–41], parameter calibration [42,43], and surrogate modelling [44–50]. These capabilities are particularly appealing for nuclear applications, where data are often sparse, costly to generate, and distributed across different reactor types and irradiation conditions [51]. By integrating diverse datasets and learning from both experimental and simulated information, ML has the potential to enhance predictive fidelity, reduce computational expense, and support the design and qualification of materials for increasingly demanding reactor environments.

Despite the rapid growth of machine learning across the broader materials science community, its application to structural materials for nuclear systems remains comparatively limited. A survey of the Web of Science highlights this disparity. Using the combined search query “machine learning” AND “material model” OR “machine learning” AND “nuclear material” (past five years, SCI index, English, article type, research areas: engineering, materials science, mechanics, metallurgy, nuclear science), a total of 9317 publications were returned, dominated by studies related to general materials modelling. When the search was restricted to the conjunction “machine learning” AND “material model” AND “nuclear materials” under the same filters, just 167 publications were obtained. Bibliometric mapping of these datasets further reveals that most ML activity clusters around alloy design, microstructure prediction, and computational materials modelling, whereas keywords related to irradiation damage, corrosion under extreme environments, and reactor structural components are far less represented. This discrepancy suggests that many advances in data-driven materials research have yet to be fully leveraged by the nuclear materials community. These observations highlight the need for a focused and critical review that consolidates existing ML efforts in nuclear materials, identifies current limitations, and highlights opportunities for integrating emerging ML methodologies into nuclear materials research.

To further illustrate this disparity, Figs. 1 and 2 show the VOSviewer [52] keyword co-occurrence maps for the two Web of Science datasets. The broader literature, given in Fig. 1, exhibits a dense, multi-cluster structure in which machine learning occupies a central position connected to diverse topics such as alloys, microstructure prediction, damage modelling, and surrogate simulation methods. In contrast, the corresponding map specifically for nuclear materials (Fig. 2) is markedly sparse, with only a few clusters and weak interconnections among keywords. These visual differences highlight the limited penetration of ML into nuclear materials research and underscore the opportunity to adapt methodologies from the broader materials community to address nuclear-specific challenges.

This review aims to address this gap by providing a comprehensive and critical assessment of machine learning applications relevant to the modelling of structural nuclear materials. We synthesise recent progress across different material classes, modelling objectives, and environmental conditions, with particular emphasis on how ML can complement or enhance existing traditional modelling approaches.

By organising recent work according to structural material class, extreme environmental conditions, and machine learning approaches developed beyond, but applicable to, structural nuclear materials, this review provides a practical, engineering-oriented perspective on the

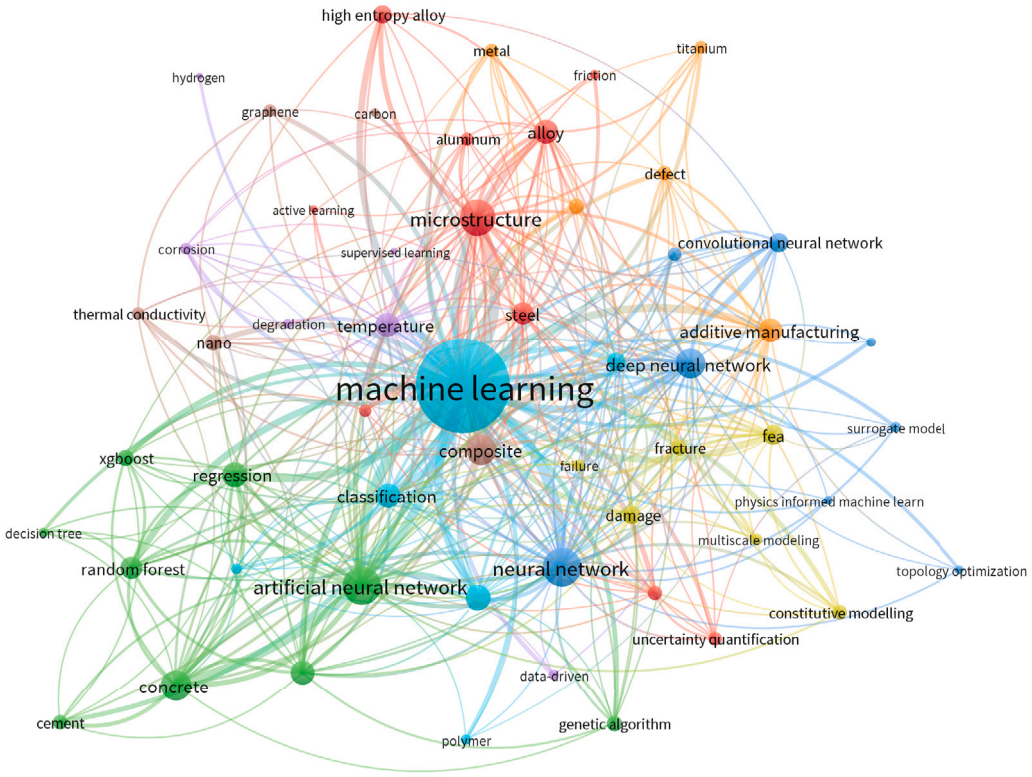


Fig. 1. VOSviewer keyword co-occurrence map for the Web of Science search "machine learning" AND "material model" OR "machine learning" AND "nuclear material" (past five years, SCI index, English, article type) [52].

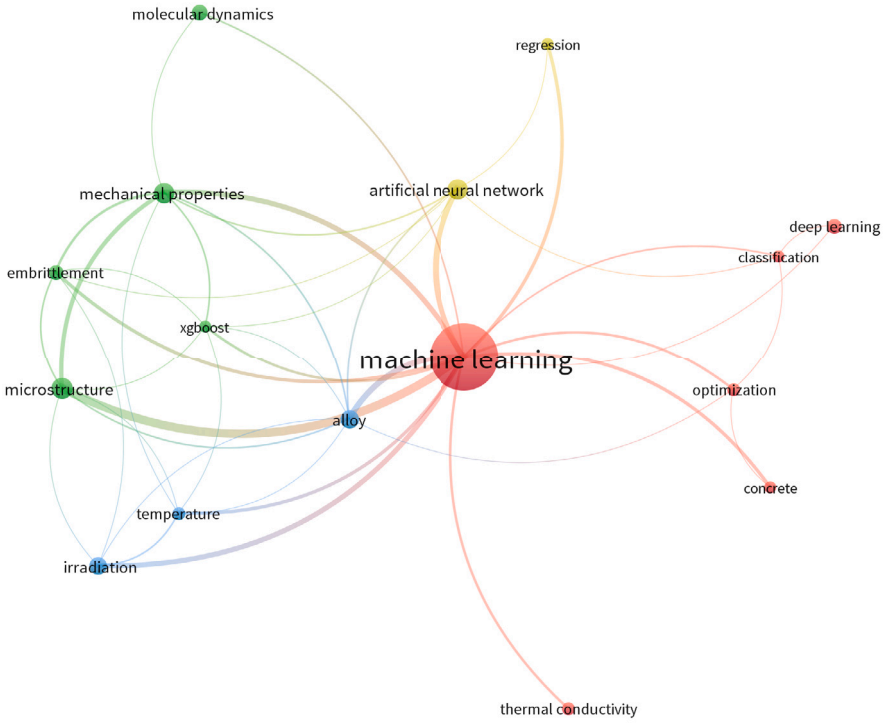


Fig. 2. VOSviewer keyword co-occurrence map for the Web of Science search "machine learning" AND "material model" AND "nuclear materials" under the same criteria as Fig. 1 [52].

current and emerging role of machine learning in modelling nuclear structural materials.

The review is structured as follows. Section 2 provides a conceptual overview of machine learning in materials modelling, introducing key modelling dimensions and representative ML architectures. Section 3 reviews ML applications across the major structural material classes used in nuclear systems, including steels, nickel-based alloys, zirconium alloys, and composite systems. Section 4 examines ML approaches for modelling degradation mechanisms under extreme environmental conditions such as irradiation, high temperatures, and corrosion. Finally, Section 5 summarises the main findings of the review and outlines future directions for integrating machine learning into nuclear materials research.

2. Machine learning in material modelling: Conceptual overview

2.1. Key modelling dimensions in ML-based material modelling

Machine learning applications in materials modelling can be organised along several conceptual dimensions that reflect the structure of the underlying physical problem and the nature of the available data. These dimensions help clarify why particular ML approaches are adopted in different modelling contexts.

2.1.1. Classification vs regression

Materials ML tasks are either classification, which assigns discrete labels such as failure mode, phase identity, or damage state, or regression, which maps inputs to continuous outputs such as stress, strain, or service lifetime. The distinction determines the loss function, output activation, and evaluation metrics. Many materials problems are naturally regression tasks, but classification is prevalent in microstructure characterisation and defect detection. For example, classification can be used to predict the phase constitution of high-entropy alloys, distinguishing between FCC, BCC, FCC + BCC, and amorphous structures from compositional descriptors [53]. Some problems combine both: a model may first classify a deformation regime before applying a regime-specific constitutive law, as demonstrated in recent surrogate modelling approaches where a classification network distinguishes elastic from elastic–plastic increments and a regression network then predicts the corresponding stress–strain evolution under multiaxial loading conditions [44,49].

2.1.2. Sequence modelling (time series) vs static snapshots

Path-dependent, rate-dependent, and irreversible material responses require models that retain memory of past states. Recurrent neural networks (RNNs), LSTMs, and temporal convolutional networks (TCNs) are designed for this purpose [54]. Where the output depends only on current state variables, or where history is adequately captured by explicit internal variables, feed-forward networks and Gaussian process regression on static snapshots are sufficient [55]. The choice has direct consequences for training complexity, interpretability, and extrapolation behaviour.

2.1.3. Processes with spatial extent vs point processes

Spatially distributed phenomena such as localised plasticity, crack propagation, and phase field evolution exhibit strong microstructure–property coupling that point-wise models cannot capture. Convolutional neural networks (CNNs), graph neural networks (GNNs), and neural operators such as the Fourier neural operator (FNO) operate on fields and learn spatially structured features [56,57]. For point processes, such as yield strength as a function of composition and temperature, standard feed-forward or Gaussian process models are appropriate.

2.1.4. Physics-based vs purely data-driven

The degree of physical prior incorporated into an ML model is a continuous choice, not a binary one. Purely data-driven models are flexible but may violate thermodynamic constraints and extrapolate poorly. Physics-informed neural networks (PINNs) and thermodynamics-based neural networks embed governing equations directly as constraints or regularisers [58,59]. Constitutively-aware neural networks (CANNs) enforce objectivity and material symmetry by construction [60]. The appropriate balance depends on data volume, the required extrapolation range, and interpretability requirements.

2.1.5. Forward prediction vs inverse design

Forward modelling predicts material response given composition, microstructure, or loading; inverse design seeks the inputs that achieve a target response. Inverse problems are substantially harder because the mapping is often non-unique and the feasible space is high-dimensional. Generative models (VAEs, GANs, diffusion models) and Bayesian optimisation are the main strategies, enabling efficient search over microstructure or composition space [61]. This distinction is important for materials discovery and process optimisation.

2.1.6. Data availability and quality

Data volume and fidelity constrain which ML strategies are viable. Large curated datasets, available through repositories such as the Materials Project [62], support expressive models with minimal physical prior. For scarce or expensive data, effective approaches include transfer learning, physics-informed regularisation, active learning, and Gaussian processes, which are sample-efficient and yield calibrated uncertainty estimates. Multi-fidelity frameworks that combine cheap low-fidelity data with sparse high-fidelity observations can substantially reduce experimental or computational cost [63].

2.2. Representative machine learning architectures for material modelling

Machine learning techniques enable the extraction of patterns and predictive relationships from data without explicit specification of governing equations. In material modelling, ML is increasingly employed as both a complement to and a substitute for traditional approaches, particularly when complex, nonlinear interactions span multiple length and time scales or when mechanistic understanding is incomplete.

Table 1 provides examples of the most widely used machine learning techniques and modelling tasks in materials science and engineering.

3. ML applications by material class

Structural nuclear materials span a wide range of metallic and ceramic systems, with material selection guided by multiple, often competing considerations. Performance requirements and long-term waste management represent two of the main competing factors. In both fusion and fission reactor systems, material options are wide-ranging. As summarised in Table 2 for fusion and in Table 3 for fission, current reactor designs rely on a broad portfolio of steels, zirconium alloys, nickel alloys, copper alloys, ceramics, and composites, reflecting both the severity of service conditions and the absence of a single universally suitable material system [87–91]. The qualification and optimisation of these materials remain major challenges, given the difficulty of generating credible laboratory-scale data and the compressed development timelines. This motivates the need for more agile approaches to material qualification for nuclear applications.

Machine learning applications in fusion and fission relevant materials are therefore discussed in this section from a materials-class-oriented perspective. The literature is reviewed according to the most commonly employed and emerging material classes, including ferritic–martensitic steels, austenitic stainless steels, nickel-based alloys, copper alloys, as well as advanced composites and other material systems.

Table 1
Representative machine learning techniques used for material modelling across different material systems.

ML technique	Task	Material	Reference
FCNN	Constitutive modelling (plasticity, damage, cyclic response)	Polycrystals; lightweight steels; metals; titanium alloys; aluminium; polymeric syntactic foam; porous materials	[44–49,60,64,65]
PINNs	Physics-constrained constitutive law discovery; Calibration	FCC metals; short-fibre reinforced polymers; digital materials; elastomers; soft materials	[66–71]
CNN/TCN	Constitutive modelling	Hyperelastic materials; rock; steel; concrete	[72–74]
RNN/LSTM	History-dependent constitutive behaviour; parameter identification	Metals; adhesives; steel; inconel	[75–79]
GPR	Stochastic constitutive modelling; uncertainty quantification	Composites; heterogeneous solids; austenitic stainless steel	[80–83]
Transformer/LLMs	Constitutive modelling	Hyperelastic materials; steel	[84–86]

Table 2
Materials used in fusion reactor components.

Component	Sub-component	Materials
First Wall	1st wall armour	Tungsten and other refractory metals; Beryllium; Molybdenum; Liquid metals; Silicon carbide; Carbon; High Entropy Alloys.
	1st wall heat sinks	P91; CuCrZr; Zirconium alloys; Graphite; DS copper*; RAFM steel*; SiC-SiC*.
	1st wall interlayer	Pure copper; Cu-W; Vanadium.
Divertor	Divertor armour	Tungsten; Molybdenum; Vanadium; Tantalum; alloy variants; Carbon/Graphite; High Entropy Alloys.
	Divertor heat exchanger	Steels; CuCrZr; Refractory metals; Zirconium alloys.
	Divertor interlayer	Pure copper; Cu-W and Cu-SiC composites; Titanium and Ti-W composites; Vanadium.
Breeder Blanket	Blanket structure	RAFM steels (Grade 92, EU97, F82H, CLAM); Advanced ferritic steels (CNA or ODS); Stainless steel (316LN); Zirconium alloys; Tungsten carbide; Tungsten; Tungsten carbide with iron chromide binder; Tungsten boride.
	Blanket shielding	Boron carbide; Boronated polyethylene; Solid methane; Concrete; Zirconium hydride; Magnesium borohydride.
	Breeder material	Li; Li_2O; Li_4SiO_4; Li_2TiO_3; Be.
	Neutron multiplier	$Be_{12}Ti$; Lead; $Li_{17}Pb_{83}$.

Table 3
Structural materials in pressurised water fission reactor [91].

Component	Sub-component	Materials
Primary Circuit	Steam Generator	Steam dryers: 304 SS; Vessel wall: Low alloy steel; Tubing: Alloys 600MA, 600TT, 690TT, 800; Divider plate: Alloy 600; Tube supports: 405 SS; Anti-vibration bars: Alloy 600, 405 SS; Primary plenum clad: 308, 309 SS.
	Reactor	Vessel: Alloy steel; Clad: 308, 309 SS; Control rod:SS clad; B_4C + SS poison; CRDM housing: Alloy 600MA, 690TT; Closure studs: Alloy steel; Core structurals: 304 SS; High strength components: A286, X750; Fuel cladding: Zry-4, advanced Zr alloys.
	Pressuriser	Vessel: Alloy steel; Clad: 308, 309 SS.
	Coolant pump	High strength: A286, 17-4 PH, X750; Structural: 304, 316 SS.
	Primary piping	304, 316 SS.
Secondary Circuit	Piping	Carbon steel.
	MSR	439 ferritic steel.
	High Pressure	Rotor: Low alloy steel; Blades: 17-4 PH, 403 SS; Blade attach:
	Steam Turbine	Low alloy steel; Diaphragm: Cr steel.
	Generator	Retaining ring: High strength, high toughness copper conductors.
	Condenser	Tubes: Ti or SS tubes; Structural: Carbon steel.
	Preheater	304 SS.

This organisational approach deliberately extends beyond studies explicitly motivated by nuclear applications, capturing machine-learning modelling efforts developed for these material classes in broader engineering and materials contexts. By doing so, it enables the nuclear materials community to identify transferable methods, insights, and data-driven techniques that may otherwise remain disconnected from nuclear research, despite their direct relevance.

3.1. Steels

Steels constitute the most extensively studied class of structural materials in the machine learning literature reviewed here, reflecting both their widespread use across engineering applications and their relevance to nuclear and fusion energy systems. Table 4 summarises ML applications in the modelling of steels (e.g., austenitic and dual-phase

Table 4
Machine learning applications in steels.

Steel type	ML type	Purpose of ML	Ref.
Austenitic SS (304L;316L;316H)	FCNN; GA;RF; GP; BO-LSTM; SVM; RL; Transformer	Predict thermal deformation behaviour and texture evolution; Predict high-temperature compressive flow behaviour; construct processing maps; Predict thickness uniformity/control in 20-high rolling of ultra-thin foil; identification of crystal plasticity model parameters; ML-informed parameter identification for visco-plastic cyclic relaxation (creep-fatigue coupling); implemented in user subroutine (UHARD)	[43,86,92–98]
Dual-phase steel (DP590;DP780;DP800)	GP; FCNN	Rate- and temperature-dependent plasticity (incl. dynamic strain ageing variable) + fracture initiation modelling; Temperature- and rate-dependent plasticity model for fracture experiment analysis; implemented in user subroutine; Identify/estimate flow curve (hardening behaviour); identification of crystal plasticity model parameters; Identify ferrite/martensite phases in microstructures; support linking phase constitutive properties to plastic heterogeneities	[95,99–102]
Duplex SS	FCNN; ANFIS	Reconstruct/predict hot-working flow stress curves from deformation test data	[103]
Advanced high-strength steel	XGBoost; DNN; PGNN	Predict bending/springback performance across geometries; Predict mechanical properties at elevated temperatures; Forming-limit assessment/FLD prediction using physics + microstructure/texture descriptors	[104–106]
Ferritic-martensitic steel	DNN	Surrogate-assisted uncertainty quantification/sensitivity for creep prediction using microstructural FE model	[107]
Low-alloy steels (42CrMo4; 12CrNi2; 2.25Cr-1Mo)	FCNN; PINN	Replace Johnson-Cook flow law with ANN; implement into Abaqus Explicit (VUHARD) and validate on necking/Taylor impact tests; Predict molten pool morphology + defect identification in DED; derive process windows from ML decision boundaries; High-temperature creep rupture life prediction using short-term data with improved extrapolation	[108–110]
Medium-carbon steel (AISI 1045)	SVR; DT; RF; FCNN	Model/predict flow stress under hot deformation as a function of strain, strain rate, temperature;	[111,112]
Lightweight steel (Fe-11Mn-6Al-0.1C)	FCNN	Predict hot flow behaviour; identify hot-working windows	[64]
Maraging steel (Fe-Ni-Al)	RF; GBR; XGBoost; SVR; DT; KRR; DNN	Predict hot-deformation flow stress; enable improved processing maps	[113]
Pipeline steel (X80; girth weld regions)	DNN	Infer true stress–strain curves/constitutive relationships from small punch test load–displacement data (with FE-generated training data)	[114]

steels, duplex stainless steels, maraging steels, advanced high-strength steels, and structural steels). The reviewed studies address diverse modelling objectives, such as deformation behaviour, constitutive parameter identification, creep and viscoplastic response, fracture and forming limits, and process optimisation, using a wide spectrum of ML techniques ranging from neural networks and Gaussian process surrogates to physics-informed and hybrid approaches. Although many of these works are not explicitly motivated by nuclear applications, the underlying steel classes, deformation mechanisms, and modelling challenges closely align with those encountered in nuclear structural materials, making these studies highly relevant as sources of transferable methodologies and insights for data-driven modelling under extreme service conditions.

Machine learning techniques have been widely employed to reproduce the constitutive behaviour and mechanical response of steels, most often using neural-network-based approaches. In some cases, these models are integrated into finite element or constitutive modelling workflows, effectively serving as surrogates. Gaussian process models and ensemble learning methods are applied mainly for parameter identification, response prediction, and process-related modelling tasks. ML has largely been adopted as a practical modelling aid within established mechanics-based frameworks. While the current body of work demonstrates clear value in improving modelling efficiency and flexibility, the development of more physically grounded and generalisable ML approaches for steels remains at a comparatively early stage.

3.2. Nickel-based alloys

Nickel-based alloys, particularly nickel-based superalloys, are widely used in high-temperature and high-load applications due to their excellent mechanical strength, creep resistance, and microstructural stability. As summarised in Table 5, machine learning has been increasingly adopted to complement physics-based modelling. Among the reviewed studies, unsupervised learning is used to extract physically meaningful low-dimensional representations of dislocation activity from in-situ diffraction data, supervised neural networks are used to correlate microstructural descriptors with mechanical properties, and hybrid ML-physics frameworks are used to accelerate crystal plasticity parameter identification. In addition, data-driven models have been applied to predict fatigue crack growth rates and high-temperature flow stress, demonstrating improved accuracy and reduced reliance on extensive experimental campaigns.

Machine learning applications in nickel-based alloys are primarily centred on linking microstructure and mechanical response. FCNN-based approaches are the most commonly applied, used to predict fatigue behaviour, hardness, and high-temperature response, as well as to support calibration of crystal plasticity models. Unsupervised methods appear in more specialised roles, such as extracting physically meaningful features from in-situ diffraction data and reducing computational cost in optimisation tasks. Overall, ML is mainly used to complement physics-based modelling by improving parameter identification and response prediction. Current studies demonstrate clear value in handling complex datasets and accelerating modelling workflows.

Table 5
Machine learning applications in nickel-based alloys.

Nickel material	ML type	Purpose of ML	Ref.
AM nickel-based superalloy	Unsupervised learning (LLE)	Analyse in-situ X-ray diffraction data to uncover low-dimensional representations of dislocation motion and defect density evolution during plastic deformation	[115]
Nickel-based superalloy	GA; SVR; FCNN; GPSR	Enable parametrically upscaled crystal plasticity constitutive modelling by mapping microstructural descriptors to constitutive coefficients in a multiscale framework	[116]
Nickel-based superalloys	FCNN	Predict fatigue crack growth rate using composition, mechanical properties, temperature, and loading parameters; Correlate gamma′ precipitate microstructural features from SEM images with hardness properties	[117,118]
Nickel alloy A230	FCNN	Predict high-temperature flow stress and outperform conventional Arrhenius-based constitutive models	[119]
AM nickel-based alloy (Haynes-214)	FCNN	Accelerate calibration of crystal plasticity parameters and predict temperature-dependent tensile behaviour	[120]

Table 6
Machine learning applications in copper.

Material	ML type	Purpose of ML	Ref.
Copper (polycrystalline)	LR; SVM; KNN; DT; RF; FCNN	Identify crystal plasticity constitutive parameters directly from macroscopic stress–strain curves; link parameters to yield stress and hardening	[42]
Copper (polycrystalline)	Symbolic regression; FCNN; SVM	Develop data-driven flow stress/constitutive models across wide strain-rate and temperature regimes; validate against impact data	[121]
Copper (polycrystalline)	FCNN	Determine active slip systems and slip magnitudes in rate-independent crystal plasticity models	[122]
Copper (single crystals)	FCNN	Eliminate loading-rate and mesh-size effects in multiscale crystal plasticity homogenisation using ML-derived constitutive relations	[123]
Copper (AM)	FCNN	Predict relative density, shrinkage, and deformation during sintering; embed ANN as constitutive law in FE simulations	[124]
Copper alloy (C19010)	ANN; GA	Accelerate parameter identification of Chaboche-type viscoplastic models describing stress relaxation behaviour	[125]

3.3. Copper alloys

As summarised in Table 6, machine learning has been applied to copper to augment or replace conventional constitutive modelling approaches across multiple length scales and deformation regimes. A dominant theme in the literature is the use of machine learning for constitutive parameter identification and model discovery within crystal plasticity and viscoplastic frameworks, enabling direct calibration from macroscopic stress–strain data or accelerating optimisation-based identification procedures. Related studies employ machine learning to determine active slip systems, perform multiscale homogenisation, and construct free-form flow stress relations spanning wide ranges of strain rate and temperature. Beyond classical deformation modelling, machine learning has also been used to describe complex functional responses in copper-containing composites and to embed data-driven constitutive laws into finite element simulations of additively manufactured copper components.

FCNNs are the most commonly used, applied to identify crystal plasticity and viscoplastic parameters, determine active slip systems, and develop flow-stress relations across wide strain-rate and temperature regimes. Ensemble techniques are applied to support constitutive model discovery and calibration from macroscopic stress–strain data. This reflects a strong emphasis on improving calibration efficiency and extending constitutive modelling capabilities, with comparatively fewer studies addressing broader materials design, uncertainty, or generalisable microstructure–property relationships.

3.4. Ceramic matrix composites

Ceramic matrix composites (CMCs) are advanced composite materials consisting of ceramic fibres, whiskers or particles embedded within a ceramic matrix. They were developed to overcome the inherent brittleness of monolithic ceramics by improving fracture toughness, while retaining high temperature stability, hardness and corrosion resistance. CMCs represent a distinct class of high-temperature structural materials in which performance is governed by complex interactions between brittle ceramic matrices, reinforcing fibres, and process-induced defects. These multi-phase architectures cause substantial challenges for conventional modelling approaches, particularly under coupled thermal, chemical, and mechanical loading conditions. As summarised in Table 7, recent work has increasingly employed machine learning to complement physics-based models across multiple length scales in CMC systems. Reported applications include data-driven constitutive modelling of fibre–matrix interphases, physics-informed prediction of creep rupture life, surrogate modelling of delamination behaviour in porous woven composites, and atomistic-scale modelling of creep mechanisms in amorphous ceramic phases relevant to composite performance.

3.5. Other materials

Beyond steels, nickel-based alloys, copper alloys, and CMC systems, the wider machine-learning literature contains additional metallic, ceramic, and structural-composite material systems (Table 8). Although these studies are often motivated by manufacturing, impact, or civil-structure performance rather than nuclear qualification, they provide transferable patterns in how ML is being used: (i) to learn constitutive relationships from mechanical test data, (ii) to accelerate parameter identification for established constitutive models, and (iii) to embed data-driven material laws into finite element simulations.

Across these diverse materials and applications, machine learning is primarily used for constitutive-response prediction, parameter identification, fracture modelling, microstructural quantification, and integration of data-driven material descriptions into finite element or atomistic simulations. Although the application contexts vary, the modelling strategies closely resemble those employed in steels, nickel alloys, copper alloys, and CMC systems discussed earlier.

Table 7
Machine learning applications in ceramic matrix composites (CMCs).

CMC material	ML type	Purpose of ML	Ref.
CMC interphase	Autoencoder, FCNN	Generate high-precision stress–strain relations for the CMC interphase under coupled thermo-chemo-mechanical loading, trained on MD simulations and calibrated with limited experiments	[126]
Oxide/oxide CMC	Physics-informed ML	Predict creep rupture life using physical constraints and microstructural features; enable high-throughput exploration of composition and modulus for design guidance	[127]
Woven CMC with porosity	LSTM	Surrogate prediction of mode I delamination load–displacement response and quantification of pore size, location, and orientation effects	[128]
SiC-based CMC	FCNN	Develop ML-based atomistic potential to simulate high-temperature compressive creep mechanisms in amorphous SiC regions relevant to CMCs	[129]

Table 8
Machine learning applications in other materials.

Material type	ML type	Purpose of ML	Ref.
Al alloys (AA7075; AA5005; 6082)	XGBoost; FCNN	Predict hot deformation flow stress behaviour and stress–strain curves; reduce experiments; Predict flow stress during hot deformation ; compare vs Arrhenius-type constitutive equation; predict non-proportional multiaxial elastic–plastic response	[44,130,131]
Titanium (Pure, Ti-6Al-4V)	GP; FCNN	Predict cyclic behaviour; Identify Johnson-Cook parameters for porous materials; Data-driven ductile fracture criterion under high-speed impact; predict ballistic resistance/simulate ballistic impact using data-driven constitutive + fracture scheme	[47,132–134]
Ni-Ti-Hf alloy	FCNN	Predict flow stress and construct processing map	[135]
Al-Zn-Mg alloy	FCNN; RF; GB; XGBoost; SVR	Predict hot-deformation flow stress ; compare ML vs Arrhenius baseline	[136]
GH4169	FCNN	Learn/predict constitutive relationships from hot-compression data and embed FCNN into FE simulations	[137]
AZ31B Mg alloy	FCNN; GA	Predict warm tensile flow stress	[138]
steel-polymer concrete panel	FCNN	Predict temperature-dependent properties of polymer concrete; modelling of steel-polymer concrete panel	[139]
Steel-concrete	FCNN; DT; GB; LightGBM; AdaBoost; voting/stacking ensembles	Identify hard-to-derive contact equation parameters for contact-stiffness correction;Predict damage/constitutive response; Predict ultimate axial load capacity of CFST columns	[140–142]
Zirconium carbide	FCNN	Simulate irradiation damage and defect evolution via molecular dynamics	[143]
Titanium-zirconium alloys	Extra Trees regression	Predict Young's modulus and design low-modulus biocompatible alloys	[144]
Improved zirconium alloy	FCNN	Estimate mechanical properties and assess cladding integrity under drop accidents	[145]
Zr-Nb-Sn alloy	Classification	Quantify recrystallisation and grain growth behaviour	[146]
Zirconium oxide composites	CART	Predict tensile strength of ABS-ZrO_2 composite filaments	[147]

The studies reviewed across steels, nickel-based alloys, copper systems, ceramic matrix composites, and additional structural materials reveal a consistent pattern in how machine learning is currently being deployed in structural materials modelling. In most cases, ML functions as an enabling layer within established mechanics and materials workflows: accelerating constitutive calibration, supporting microstructure–property mapping, or providing surrogate representations for computationally intensive simulations. Far fewer studies attempt to replace physics-based descriptions entirely; instead, data-driven methods are typically embedded within existing modelling hierarchies.

This reflects both the opportunities and constraints associated with structural materials research. The strongest progress is observed where rich experimental datasets and mature modelling frameworks already exist, allowing ML to enhance efficiency. In contrast, in emerging materials systems, ML is more frequently used to explore composition–processing–property spaces and to support early-stage materials discovery. The balance between these two roles is particularly significant in nuclear materials research, where legacy alloys, regulatory requirements, and long qualification timelines shape how modelling advances can be translated into practice.

Nuclear industry is currently going through a regeneration. This is because of the geopolitical events of the recent years combined with the impacts of the climate emergency. While there is a plan to build plants at speed to tackle the energy sovereignty, which inevitably limits the scope for using new materials as their adoption by the industry due to regulatory restrictions can be limited, there is also a demand to reduce the cost of nuclear and enhance their lifetime which motivates the use of materials. In this sense, the application of machine learning to nuclear materials could perhaps be split into two categories. One is applying machine learning to materials the nuclear industry has been using for decades where there are large experimental datasets at various length scales and established computational, albeit slow, modelling techniques (often fission) and the other is a more future-looking application of machine learning to develop new alloys (often fusion). An example of the former is the use of machine

learning to bridge the length scales between models and to combine experimental and modelling techniques to reduce unnecessary over-conservatism in ubiquitous alloys such as austenitic stainless steels or ferritic-martensitic alloys. An example of the latter is the rapid discovery of high-entropy alloys for Gen IV and fusion reactors to withstand harsher environments. The use of machine learning to assist with our available mechanistic understanding allows seemingly small yet consequential steps in the better use of accepted alloys within the current regulatory framework to extend the life of the current systems, thus reducing their cost over their lifetime while enabling researchers and operators to work with regulators to deploy new long-enduring alloys in future energy systems safely.

4. ML applications for extreme environmental effects

Structural materials in nuclear reactors experience a complex combination of extreme environmental conditions, such as elevated temperatures, intense neutron and ion irradiation, and corrosive chemical environments. These conditions induce coupled degradation mechanisms that are difficult to predict using traditional physics-based models alone. Machine learning offers a powerful complementary approach, with the ability to capture complex nonlinear relationships among material composition, processing history, microstructure, environmental parameters, and degradation responses. This section reviews ML applications for modelling material behaviour under three primary extreme conditions: high temperature, corrosion, and irradiation, as well as their synergistic interactions.

4.1. High temperature

Thermal creep, dynamic recrystallisation, oxidation, and thermally activated phase transformations are the main degradation mechanisms that occur under high-temperature exposure. Accurate prediction of these behaviours is essential for component lifetime assessment and safe reactor operation. Table 9 provides a structured overview of recent ML-enabled studies addressing material behaviour under high-temperature conditions.

The reviewed works span a wide range of material classes, from metallic alloys to composites, polymers, and elastomers, and encompass diverse thermomechanical regimes such as hot deformation, creep-fatigue interaction, cyclic plasticity, and temperature-dependent flow stress. Across these studies, several ML techniques are used either as standalone surrogates or in hybrid formulations coupled with established constitutive models.

4.1.1. Creep life prediction

Creep rupture life prediction is one of the most critical applications of ML in high-temperature nuclear materials research. Traditional empirical methods such as the Larson-Miller parameter, Monkman-Grant relationship, and θ projection method suffer from oversimplified assumptions, poor generalisability, and inability to capture complex nonlinear relationships [174]. Machine learning addresses these limitations by simultaneously handling many input features, learning nonlinear interactions, and demonstrating robust generalisability.

Recent reviews group machine learning approaches for creep life prediction into four main categories: (i) hybrid modelling approaches that combine ML with physics-based models, (ii) creep mechanism-aware models that incorporate domain knowledge, (iii) feature selection techniques that improve interpretability, and (iv) uncertainty quantification methods for safety-critical applications [174]. Physics-informed neural networks (PINNs) have shown particular promise for extrapolating creep rupture life from short-term data to long-term predictions, addressing a fundamental challenge in nuclear materials qualification [110].

4.1.2. Hot deformation and constitutive modelling

Machine learning has been extensively applied to predict hot deformation behaviour and construct constitutive models for materials operating at elevated temperatures. Studies on austenitic stainless steels [93,97], titanium alloys [48,153–157], and nickel-based superalloys [119,120] demonstrate that ML models can outperform traditional Arrhenius-type constitutive equations in predicting flow stress as a function of strain, strain rate, and temperature.

Physics-informed machine learning approaches are particularly effective for hot deformation modelling. By embedding physical constraints and metallurgical principles directly into the learning framework, these models achieve improved extrapolation capabilities while maintaining physical interpretability [150,153]. For example, physics-guided ML approaches that incorporate knowledge of dynamic recrystallisation mechanisms have successfully predicted both flow stress behaviour and microstructural evolution during thermomechanical processing [157,161].

For aluminium alloys, a diverse array of ML techniques (PINNs, FCNN, and XGBoost) have been employed to predict rheological behaviour under coupled temperature, strain, and strain-rate conditions during hot deformation [130,131,150,151]. These studies consistently demonstrate superior accuracy compared to Arrhenius-based constitutive models, particularly in capturing the complex nonlinearities associated with dynamic recrystallisation and strain softening.

Copper-based materials have also received considerable attention. Symbolic regression approaches have been used to develop interpretable data-driven flow stress models across wide temperature and strain-rate regimes [121]. For antibacterial stainless steels containing copper, physical metallurgy-guided ML combining least squares support vector machines with particle swarm optimisation has proven effective for predicting hot deformation mechanisms [98].

High-entropy alloys (HEAs) represent an emerging class of materials for extreme environments, and ML techniques such as Black Kite Algorithm-SVR, LSTM networks, and interpretable XGBoost models have been applied to predict their high-temperature flow stress, dynamic recrystallisation behaviour, and processing maps [161–163].

4.1.3. Cyclic and creep-fatigue behaviour

High-temperature cyclic loading and creep-fatigue interactions pose significant challenges for nuclear structural materials, particularly in components subjected to thermal transients. ML-based approaches have been developed to model viscoplastic cyclic relaxation [96], temperature-dependent cyclic plasticity [48], and combined creep-fatigue behaviour [149].

Gradient-based optimisation with automatic differentiation has been employed to calibrate constitutive model parameters against high-temperature creep-fatigue experimental data for general high-temperature structural materials [148]. For 316H stainless steel, Gaussian process regression combined with Bayesian inference has been used to identify viscoplastic parameters governing cyclic relaxation behaviour at 550 °C [96].

Surrogate models trained on physics-based simulations allow rapid assessment of creep-fatigue reliability at elevated temperatures [149]. These data-driven roadmaps significantly accelerate the typically time-consuming process of creep-fatigue life assessment, thereby enabling more efficient qualification of materials for demanding reactor environments.

4.1.4. Composites and advanced materials

Machine learning applications extend beyond metallic alloys to a wide range of composite and polymeric materials relevant to nuclear applications. For carbon fibre-reinforced polymer composites, Gaussian process regression has been employed to model temperature-dependent shear nonlinearity and to integrate the predictions into finite element simulations [164].

Table 9
Machine learning applications for high-temperature behaviour prediction in nuclear-relevant materials.

Material/System	ML technique	Purpose of ML	Ref.
Creep and Creep-Fatigue			
High-temperature structural materials (general)	Gradient-based optimisation	Calibrate constitutive model parameters against high-temperature creep-fatigue experimental data	[148]
2.25Cr-1Mo steel (Grade 22)	PINN	Predict long-term high-temperature creep rupture life with improved extrapolation	[110]
316H SS	GP + BI	Identify viscoplastic parameters for cyclic relaxation at 550 °C	[96]
Low-pressure turbine disk material	SVR; GRNN	Accelerate creep-fatigue reliability assessment at elevated temperatures	[149]
Hot Deformation and Flow Stress - Aluminium Alloys			
AA6061-O Al alloy	PINN	Predict rheological behaviour under coupled temperature, strain, and strain-rate conditions during hot deformation	[150]
AA5005 Al alloy	FCNN	Predict high-temperature flow stress and outperform Arrhenius-based models	[131]
AA7075 Al alloy	XGBoost	Predict hot deformation flow stress and entire stress–strain curves	[130]
AA5052-H32 Al alloy	FCNN; XGBoost	Predict hot deformation flow behaviour across temperatures and strain rates	[151]
Hot Deformation and Flow Stress - Steels			
Medium-carbon steel (AISI 1045)	SVR; DT	Model hot deformation flow stress as a function of strain, strain rate, and temperature	[111]
P91 steel	BO	Automated calibration of ductile damage parameters at elevated temperatures	[152]
022Cr19Ni10 austenitic SS	BPNN; PSO-BPNN; GA-BPNN	Predict high-temperature flow stress and establish hot processing maps	[97]
304L SS	RF; ANN	Predict high-strain-rate, high-temperature flow behaviour and construct processing maps	[93]
High-strength steel	FCNN	Predict elevated-temperature mechanical properties from temperature and chemical composition	[105]
Dual-phase steel (DP780)	FCNN	Model rate- and temperature-dependent plasticity and fracture initiation	[99]
DP800 dual-phase steel	FCNN	Model temperature- and rate-dependent plasticity and fracture	[100]
42CrMo4 steel	FCNN	Replace Johnson-Cook flow law in FE simulations with temperature dependence	[108]
Hot Deformation and Flow Stress - Titanium Alloys			
Ti-6Al-4V	FCNN	Surrogate modelling of temperature-dependent cyclic plasticity behaviour	[48]
Ti-3Al-8V-6Cr-4Mo-4Zr	PINN	Predict strain-rate- and temperature-dependent elastoplastic response	[153]
Ti-6Al-4V	SSA-optimised FCNN	Identify temperature-dependent constitutive and friction parameters	[154]
HIPed Ti4522XD	XGB; FCNN	Predict hot deformation behaviour and DRX mechanisms	[155]
Ti-10V-5Al-2.5Fe-0.1B	Phenomenological + ML hybrid	Predict high-temperature rheological stress and construct processing maps	[156]
Titanium alloy (general)	FCNN	Predict hot deformation behaviour and DRX mechanisms	[157]
Hot Deformation and Flow Stress - Copper and Nickel Alloys			
Pure copper	Symbolic regression	Develop data-driven flow stress constitutive models across wide temperature and strain-rate regimes	[121]
Cu-containing antibacterial SS	PM-guided LSSVM	Predict hot deformation mechanisms and rheological behaviour	[98]
Nickel alloy A230	ANN	Predict high-temperature flow stress more accurately than Arrhenius models	[119]
Haynes-214 nickel-based superalloy (AM)	DT; FCNN	Predict temperature-dependent tensile behaviour and calibrate CP parameters	[120]
Hot Deformation and Flow Stress - Other Metallic Alloys			
56Ni-32Ti-12Hf alloy	CIWOA-optimised FCNN	Predict hot deformation flow stress and construct processing maps	[135]
$L1_2$-strengthened $Co_{33}Cr_{23}Ni_{34}Al_5Ti_5$ alloy	XGBoost	Predict hot deformation behaviour and optimise thermomechanical processing windows	[158]
BN-reinforced AZ80 magnesium composite	SVR	Predict flow stress and optimise thermal deformation conditions	[159]

(*continued on next page*)

Table 9 (*continued*).

AZ31 magnesium alloy (pre-deformed)	GA-optimised ANN	Predict hot shear-compression flow behaviour and workability	[160]
AZ31B magnesium alloy	FCNN; GA-BP-ANN; constrained ANN	Predict warm deformation flow stress	[138]
Hot Deformation and Flow Stress - High-Entropy Alloys			
High-entropy alloys (HEA)	Black Kite Algorithm-SVR	Predict hot deformation flow stress and DRX behaviour	[161]
NiAlCrFeMo high-entropy alloy	LSTM	Predict hot deformation behaviour and establish processing maps	[162]
HEAs for extreme environments	XGBoost and interpretable ML	Predict high-temperature and high-strain-rate mechanical properties	[163]
Composites and Polymeric Materials			
T800 carbon fibre/epoxy plain-woven composite	GPR	Model and predict temperature-dependent shear nonlinearity and integrate into FE simulations	[164]
Fibre-reinforced polymers	GRU-based RNN; CNN	Multiscale thermo-mechanical constitutive modelling in FE simulations	[165,166]
Polymeric syntactic foam	FCNN	Rapid prediction of temperature-dependent stress–strain curves	[46]
Ceramic matrix composites (CMC)	Autoen-coder+FCNN	Generate thermo-chemo-mechanical strain-stress relations of CMC interphase	[126]
CMC (creep-fatigue)	Physics-informed RNN	Surrogate modelling of time-dependent high-temperature creep-fatigue behaviour	[167]
Magnetoelectric composite plates	Multiple ML methods	Predict temperature- and moisture-dependent coupled vibration frequencies	[168]
Polymer concrete	FCNN	Predict temperature-dependent mechanical and thermal properties	[139]
Thermoplastic polymers (PEEK, PC)	Multi-fidelity Kriging-based ML	Predict temperature- and strain-rate-dependent stress–strain relationships	[169]
Elastomers and Shape Memory Alloys			
Elastomers (filled and cross-linked)	CondNN	Predict strain-rate-, temperature-, and morphology-dependent constitutive and failure behaviour	[170]
Thermo-oxidatively aged elastomers	PINNs	Predict temperature-dependent ageing-induced constitutive degradation	[171]
Shape memory alloy (SMA) wires	CNN + BiLSTM + attention (CBLS)	Predict stress and strain for adaptive high-temperature monitoring	[172]
Crystal Plasticity and Multiscale Modelling			
Polycrystalline materials (general)	PINN + K-means clustering	Accelerate FFT-based crystal plasticity simulations under complex thermo-mechanical loading	[173]

Ceramic matrix composites (CMCs) present particular challenges due to their complex interphase structures. Autoencoder architectures combined with feedforward neural networks have been used to generate thermo-chemo-mechanical strain-stress relations for CMC interphases [126]. Physics-informed recurrent neural networks have also been used for surrogate modelling of time-dependent high-temperature creep-fatigue behaviour in these materials [167].

For fibre-reinforced polymers, GRU-based recurrent neural networks and convolutional neural networks have been developed for multiscale thermo-mechanical constitutive modelling in finite element simulations [165,166]. Thermoplastic polymers have been addressed using multi-fidelity Kriging-based ML to predict temperature- and strain-rate-dependent stress–strain relationships [169].

Elastomeric materials have been modelled using conditional neural networks to predict strain-rate-, temperature-, and morphology-dependent constitutive and failure behaviour [170]. For thermo-oxidatively aged elastomers, physics-informed multi-agent neural networks have been developed to predict temperature-dependent ageing-induced constitutive degradation [171].

4.2. Corrosion

Corrosion and oxidation represent major life-limiting degradation mechanisms for structural materials operating in aggressive nuclear environments, particularly where chemical exposure is coupled with thermal, mechanical, or electrochemical driving forces. Recent machine learning applications in this area span a wide range of material systems and address corrosion phenomena both through direct predictive modelling and through surrogate-assisted alloy design and degradation-informed constitutive analysis (see Table 10).

4.2.1. Stress corrosion cracking prediction

Stress corrosion cracking (SCC) is a critical degradation mechanism affecting stainless steels in the high-temperature, high-pressure water environments of pressurised water reactors (PWRs). The SCC crack growth rate (CGR) is influenced by complex interactions among material properties, stress states, and environmental conditions, making it an ideal candidate for ML-based prediction [175].

Recent studies have employed multiple ML algorithms (Random Forest, XGBoost, Support Vector Regression, and Gaussian Process Regression) to predict CGR in 304SS and 316SS stainless steels under simulated PWR primary water conditions [175]. SHAP (SHapley Additive exPlanations) analysis has been used to interpret model predictions and identify the most influential factors affecting SCC behaviour, such as cold work level, temperature, and dissolved hydrogen concentration. These ML-assisted insights provide valuable guidance for material selection and water chemistry optimisation in nuclear power plants.

4.2.2. High-temperature and Molten salt corrosion

The development of advanced reactor concepts, notably molten salt reactors (MSRs) and high-temperature gas-cooled reactors, has intensified interest in predicting corrosion behaviour under extreme thermal and chemical conditions. Machine learning methods have been applied to estimate corrosion rates of structural materials and coatings in high-temperature environments relevant to nuclear thermochemical cycles [176].

For molten salt applications, ML and data science techniques are being developed to address the challenge of predicting long-term corrosion rates from sparse short-term data. These methods must consider sensitivity to unknowns and uncertainties to provide acceptable safety margins under both normal operation and beyond-design-basis accident conditions.

4.2.3. Zirconium alloy corrosion

Zirconium alloys serve as fuel cladding materials in light water reactors due to their favourable neutronics and mechanical properties, but their long-term performance is limited by accelerated aqueous corrosion that exhibits a kinetic transition when the oxide film reaches a critical thickness [177]. Despite decades of research, the mechanisms controlling this transition remain incompletely understood.

A recent comprehensive review has assessed the potential of advanced ML methods for enhancing understanding of oxidation and corrosion transitions in zirconium alloys under nuclear reactor conditions [177]. The review proposes an integrated framework that combines physics-based models with data-driven approaches, leveraging regression techniques, deep learning architectures, and physics-informed neural networks to accelerate the discovery of corrosion-resistant alloys and enable accurate lifetime predictions.

4.2.4. Galvanic and localised corrosion

Machine learning surrogates have been used to accelerate finite element method (FEM)-based corrosion simulations for galvanic couples relevant to nuclear applications [178,179]. AdaBoosted decision tree surrogates can replace computationally expensive FEM simulations to rapidly predict galvanic corrosion current as a function of environmental and geometric parameters [178]. Active learning-assisted approaches further minimise the required training data by strategically selecting the most informative simulations [179].

Physics-informed ML combined with uncertainty quantification has been applied to quantify galvanic corrosion behaviour and the reliability of dissimilar Al-Fe joints under environmental uncertainty [180]. For alkali-activated slag concrete, extended support vector regression metamodels have been developed to accelerate probabilistic prediction of chloride-induced corrosion initiation by replacing computationally expensive multi-ionic transport simulations [181].

4.2.5. Nuclear waste storage and corrosion

For uranium and uranium alloys, LSTM neural networks combined with particle swarm optimisation and image clustering have been employed to screen and design corrosion-surrogate alloys with oxidation behaviour similar to uranium alloys [182]. This approach enables safer and more efficient testing of corrosion behaviour without the complications associated with handling radioactive materials.

4.3. Irradiation

Irradiation environments introduce profound changes to material microstructure and properties through displacement damage, transmutation reactions, and their coupling with thermal and mechanical effects. The resulting degradation in the form of hardening, embrittlement, void swelling, irradiation creep, radiation-induced segregation (RIS), and phase instabilities evolves over timescales ranging from picoseconds to decades and across length scales from atomic to macroscopic. Machine learning has emerged as a powerful tool for accelerating both fundamental understanding and predictive capability, as illustrated by recent studies summarized in Table 11.

4.3.1. Irradiation embrittlement and hardening

Reactor pressure vessel (RPV) steel embrittlement due to neutron irradiation is a critical concern for nuclear plant life extension. The shift in ductile-to-brittle transition temperature (DBTT) and associated hardening must be reliably predicted for fluences and flux conditions beyond the current operational experience [185].

Machine learning models trained on comprehensive irradiation databases have demonstrated superior extrapolation capabilities compared to existing empirical embrittlement models. A landmark study showed that ML models can capture intermediate flux effects at high fluence, providing predictions that are more accurate than traditional models like EONY for life-extension conditions [185]. The successful extrapolation to low-flux, high-fluence conditions, characteristic of LWR extended operation, highlights the potential of ML to support reactor lifetime decisions.

4.3.2. Void swelling

Radiation-induced void swelling is a serious mode of degradation in nuclear structural materials, causing dimensional instability and potentially compromising component integrity. Machine learning has been applied to predict the onset of void swelling based on material composition and irradiation parameters [186]. Gradient boosting ensemble methods have successfully predicted experimental swelling onset doses and identified the contributions of temperature, Fe and Cr content, and dose rate consistent with established physical understanding.

This pioneering work demonstrates the feasibility of ML in predicting macroscale radiation effects from material and environmental parameters, with practical significance for guiding material optimisation [186]. More recent studies have extended these approaches to high-entropy alloys (HEAs), which show promise for enhanced swelling resistance due to their complex compositional and microstructural characteristics [187].

4.3.3. Machine-learned interatomic potentials

A transformative development in irradiation damage modelling has been the emergence of machine-learned interatomic potentials (MLIPs), which enable atomistic simulations with near-DFT accuracy at a fraction of the computational cost. These potentials are particularly valuable for radiation cascade simulations that require accurate description of the highly non-equilibrium atomic configurations produced during collision events.

ML-based surrogate formulations have also been developed to replace analytical sink-strength expressions in rate theory with surrogate models that capture nonlinear defect-sink interactions and long-term irradiation-induced bubble evolution [188].

4.3.4. Defect characterisation and microstructure analysis

Machine learning, particularly convolutional neural networks (CNNs), has revolutionised the analysis of irradiated material microstructures obtained from transmission electron microscopy (TEM) and other characterisation techniques [189,190]. Automated detection and quantification of radiation-induced defects (e.g. helium bubbles, dislocation loops, voids, and precipitates) enables high-throughput analysis that would be prohibitively time-consuming with manual methods.

A novel Generative Convergence Model (GCM) framework has been developed for unsupervised segmentation of helium bubble images in additively manufactured 304L stainless steel [189]. This approach achieves several hundred-fold speedup compared to manual analysis while significantly improving accuracy. This allows rapid assessment of irradiation performance for materials fabricated through advanced manufacturing routes.

Table 10
Machine learning applications for corrosion prediction in nuclear-relevant materials.

Material/System	ML technique	Purpose of ML	Ref.
Stress Corrosion Cracking			
316SS, 304SS	RF; XGBoost; SVR; GPR + SHAP	SCC crack growth rate prediction in PWR environment	[175]
High-Temperature and Molten Salt Corrosion			
Structural alloys and coatings	RF	High-temperature corrosion rate estimation	[176]
Antibacterial SS with varying Cu content	Physical metallurgy-guided ML (PSO-optimised LSSVM)	Predict thermal deformation mechanisms and dynamic recrystallisation; guidance for corrosion-resistant SS design	[98]
Zirconium Alloy Corrosion			
Zirconium alloys	Regression; DL ; PINNs	Corrosion transition mechanisms and lifetime prediction	[177]
Galvanic and Localised Corrosion			
Al-SS galvanic couple	AdaBoosted DT	Replace FEM-based galvanic corrosion simulations to predict corrosion current	[178]
AA7075-SS304 galvanic couple	Active learning-assisted surrogate model	Rapidly assess galvanic corrosion current using actively trained ML surrogate	[179]
Al-Fe dissimilar material joints	Physics-informed GP	Quantify galvanic corrosion behaviour and reliability under environmental uncertainty	[180]
Alkali-activated slag (AAS) concrete	Extended SVR	Accelerate probabilistic prediction of chloride-induced corrosion initiation	[181]
Nuclear Waste Storage and Uranium Alloys			
U-2.5Nb uranium alloy and surrogate alloys	LSTM ; clustering	Screen and design corrosion-surrogate alloys with oxidation behaviour similar to uranium alloys	[182]
Alloy Design for Corrosion Resistance			
Magnesium alloys	Multiple ML methods (survey)	ML-assisted alloy design, processing optimisation, and prediction of corrosion properties	[183]
Al-Mg-Zn-Er-Zr alloy	FCNN	Predict hot deformation flow behaviour; link deformation-induced microstructure to improved corrosion performance	[184]

4.3.5. High-entropy alloys for irradiation environments

High-entropy alloys are promising candidates for extreme irradiation environments due to their exceptional mechanical properties and potentially enhanced radiation tolerance [187,191]. Machine learning interatomic potentials (MLIPs) for atomistic simulation and knowledge-enabled ML for composition optimisation are accelerating the discovery and understanding of radiation-resistant HEA compositions.

ML models trained on both experimental and simulation datasets have successfully predicted phase stability, mechanical properties, and radiation damage in HEAs [187]. ML methods are enabling faster identification of HEA compositions with optimised performance under extreme conditions, opening new directions in alloy design for nuclear applications. Studies combining XGBoost with SHAP interpretability have identified key physical parameters and compositional trends affecting dynamic mechanical properties under extreme conditions [163].

4.3.6. Processing optimisation under irradiation

Bayesian optimisation surrogate models have been applied to optimise irradiation processing parameters (laser power, time, gas, and pressure) to tailor electrical and structural properties of materials such as graphene-based TiO_2 thin films [192]. This approach accelerates material optimisation under irradiation conditions by efficiently exploring the parameter space.

LSTM stacked-ensemble surrogate models have been developed for rapid prediction of irradiation-driven precipitation kinetics and the resulting mechanical property degradation in 316L stainless steel nuclear cladding [193]. These models, informed by kinetic Monte Carlo and finite element method simulations, enable efficient assessment of cladding performance evolution during service.

4.4. Coupled and synergistic effects

In real nuclear reactor environments, materials are simultaneously exposed to high temperature, irradiation, and corrosive media. These extreme conditions interact synergistically to produce degradation behaviours that are more severe than the sum of the individual effects. Understanding and predicting these coupled phenomena represent one of the greatest challenges in nuclear materials science and an emerging frontier for machine learning applications. Machine-learning-based surrogate models are particularly well-suited to capturing such coupled, nonlinear interactions, offering substantial gains in both predictive accuracy and computational efficiency compared to traditional modelling strategies.

4.4.1. Irradiation-assisted stress corrosion cracking

Irradiation-assisted stress corrosion cracking (IASCC) exemplifies the complexity of coupled degradation, involving radiation-induced segregation (RIS), irradiation hardening, and environmental interactions with the coolant [51]. The phenomenon can require approximately 10 dpa and decadal timescales to manifest for in-core components, making experimental characterisation extremely challenging and time-consuming.

Machine learning approaches that integrate information from multiple degradation mechanisms show promise for more comprehensive IASCC prediction frameworks. However, this remains an active area of research, and there is still considerable room for methodological development.

Table 11
Machine learning applications for irradiation effects prediction in nuclear materials.

Material/System	ML technique	Purpose of ML	Ref.
Embrittlement and Hardening			
RPV steels	GKRR	Irradiation embrittlement prediction for life extension	[185]
Void Swelling and Dimensional Changes			
Austenitic SS	GB	Void swelling onset dose prediction	[186]
316L SS cladding	LSTM	Rapid prediction of irradiation-driven precipitation kinetics and mechanical property degradation	[193]
Machine-Learned Interatomic Potentials and Surrogates			
Irradiated crystalline materials	GPR	Replace analytical sink-strength expressions in rate theory to capture nonlinear defect-sink interactions	[188]
Defect Characterisation			
AM 304L SS	Generative Convergence Model (GCM)	Automated helium bubble segmentation in TEM images	[189]
High-Entropy Alloys			
HEAs for nuclear applications	Multiple ML methods (survey)	Phase stability, mechanical properties, radiation damage	[187]
High-entropy materials	MLIP; ML-informed defect evolution	Irradiation damage mechanisms and predictions	[191]
HEAs for extreme environments	XGBoost, SHAP, feature screening	Design HEAs with favourable dynamic mechanical properties under extreme conditions- irradiation	[163]
Processing Optimisation			
Graphene-based TiO_2 thin films (laser-irradiated)	BO	Optimise irradiation processing parameters to tailor electrical and structural properties	[192]

4.4.2. Integration of multiple degradation modes

The comprehensive modelling of nuclear material degradation requires the consideration of multiple phenomena occurring simultaneously. Physics-informed ML approaches that embed mechanistic understanding of individual processes while learning their interactions from data offer a promising path forward [190]. A recent systematic review of ML applications in nuclear materials at the continuum scale emphasises the importance of hybrid ML-physics frameworks that can integrate with finite element simulations while ensuring that predictions remain physically meaningful [190].

4.5. Challenges and future directions

Despite significant progress, several challenges remain in applying ML to predict material behaviour under extreme environmental conditions in nuclear applications:

1. **Data scarcity and quality**: High-quality irradiation data remains expensive and time-consuming to generate, particularly for neutron irradiations at prototypic flux and fluence conditions. Transfer learning and multi-fidelity approaches that leverage ion irradiation data or computational simulations to supplement limited neutron data represent promising directions [51].
2. **Extrapolation to extended life conditions**: Nuclear materials must often be qualified for conditions (fluence, time, temperature) beyond available experimental data. Physics-informed approaches that embed mechanistic constraints show improved extrapolation capability compared to purely data-driven models [110].
3. **Uncertainty quantification**: Safety-critical nuclear applications require robust uncertainty estimates. Bayesian approaches, ensemble methods, and conformal prediction techniques are being developed to provide reliable uncertainty bounds on ML predictions.
4. **Interpretability**: "Black-box" ML predictions are insufficient for nuclear licensing where understanding of degradation mechanisms is required. Explainable AI methods such as SHAP, attention mechanisms, and symbolic regression are increasingly being integrated into nuclear materials ML workflows.
5. **Coupled multi-physics modelling**: Real reactor environments involve simultaneous exposure to multiple extreme conditions. Developing ML frameworks that can capture synergistic interactions while maintaining physical consistency remains an open challenge.

The studies reviewed show that ML-based approaches can provide clear advantages over conventional physics-based and empirical models, particularly in capturing high-dimensional, non-linear couplings among composition, microstructure, temperature, mechanical loading, irradiation dose, and environmental chemistry. ML surrogates can significantly reduce the computational cost of multiscale simulations and allow rapid parametric exploration, inverse parameter identification, and uncertainty quantification. At the same time, physics-informed and hybrid ML frameworks help maintain consistency with the underlying physical mechanisms.

Despite these advances, several important challenges remain. Current approaches are still constrained by limited availability of high-quality data, difficulties in extrapolating beyond calibrated regimes, reduced interpretability of complex model architectures, and incomplete representation of fully coupled thermo-chemo-mechanical–irradiation interactions under service conditions. Addressing these limitations will require the development of uncertainty-aware modelling frameworks, the use of multi-fidelity and transfer learning strategies for sparse long-term degradation data, and closer integration of ML with mechanistic modelling and in situ experimentation. Such efforts will be essential for progressing towards predictive, system-level life assessment of materials operating in extreme environments.

5. Conclusions

This review has examined the growing role of machine learning in understanding and predicting the behaviour of nuclear structural

materials. Across different material classes, including steels, nickel-based alloys, copper alloys, ceramic matrix composites, and other structural materials, ML is most commonly used to support established modelling approaches rather than to replace them. In many cases, ML methods serve as efficient tools for identifying constitutive parameters, mapping microstructure–property relationships, and developing surrogate models. Neural-network-based approaches remain the most widely used, although Gaussian process models, ensemble methods, symbolic regression, and physics-informed architectures are also common. The literature shows that ML can significantly accelerate modelling workflows, reduce reliance on extensive experimental campaigns, and enable more efficient exploration of parameter spaces.

When applied to extreme environmental conditions relevant to nuclear reactors, ML demonstrates particular promise in addressing degradation phenomena that are difficult to capture using conventional models alone. Examples include the prediction of creep rupture life at high temperatures, modelling of hot deformation behaviour, stress corrosion cracking prediction, irradiation-induced embrittlement, void swelling, and the analysis of radiation-induced microstructures. ML techniques have also enabled advances in areas such as machine-learned interatomic potentials for radiation damage simulations, automated defect detection in microscopy data, and surrogate models for corrosion and irradiation-driven microstructural evolution. These developments illustrate the ability of ML to integrate diverse data sources, ranging from atomistic simulations to experimental characterisation and continuum-scale modelling.

Despite these advances, several challenges continue to limit the widespread adoption of ML in nuclear materials research. Data availability remains a central constraint, particularly for irradiation experiments and long-term degradation processes that require decades of service exposure to reproduce. In addition, data-driven models often struggle to extrapolate beyond the conditions represented in their training datasets, which is problematic for nuclear applications where materials must be qualified for extended lifetimes and extreme operating environments. Ensuring physical consistency, interpretability, and robust uncertainty quantification is therefore essential for translating ML predictions into practice and regulatory decision-making.

The studies reviewed here demonstrate that ML is emerging as a valuable complement to traditional modelling and experimental approaches in nuclear materials science. ML is effective when used to enhance existing frameworks, accelerate simulation workflows, and enable new data-driven insight across multiple length and time scales. As computational and experimental capabilities evolve, the integration of ML methodologies is expected to play an increasingly important role in the development, qualification, and lifetime management of structural nuclear materials.

CRediT authorship contribution statement

Burcu Tasdemir: Writing – review & editing, Writing – original draft, Conceptualization, Funding acquisition. **Nicolas Larrosa:** Writing – original draft. **Pratheek Shanthraj:** Funding acquisition. **Mahmoud Mostafavi:** Writing – review & editing, Funding acquisition. **David Barton:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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